Anthony John Stewart

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Education

University of Washington, Seattle, WA, Advisors – Dr. Moskal and Dr. Butman 2025
Ph.D. School of Environmental and Forest Sciences (SEFS)

University of New Hampshire, Durham, NH, Advisor – Dr. Asbjornsen 2017
M.S. in Natural Resources

Montana State University, Bozeman, MT 2014
B.S. in Environmental Science, focus in Biology

Research & Professional Experience

Atkinson Postdoctoral Fellow 2025 – Present
Leading research on development of new wetland maps through deep learning to further uncover the natural climate solution potential of inland wetlands in New York State in collaboration with The Nature Conservancy and The Environmental Defense Fund

Graduate Research Assistant, UW, Seattle, WA 2020 – 2025
Developed novel research methods and tools for improving monitoring and mapping of wetlands and soil carbon with remote sensing and machine learning.

DOE SCGSR Fellow, Lawrence Livermore National Laboratory, Livermore, CA 2024 – 2025
Conducted soil carbon cycling laboratory analysis to further understanding of soil carbon persistence in wetlands and uplands using radiocarbon dating and mass spectrometry.

Laboratory Manager, Cornell University, Ithaca, NY 2017 – 2020
Managed forest ecosystem ecology research projects by operating analytical laboratory equipment and leading field research campaigns.

Graduate Research Assistant, UNH, Durham, NH 2015 – 2017
Evaluated effects of forest-to-agriculture land use change on soil hydrology providing key local scale management information.

Teaching Experience

Guest Lectures 2023 – 2025
ESRM 410: Forest Soils and Site Productivity
ESRM 430: Remote Sensing

Graduate Teaching Assistant, UW, Seattle, WA 2022 – 2024
ESRM 433: LiDAR Remote Sensing (Lead Instructor for Laboratories)
ESRM 201: Sustaining Pacific Northwest Ecosystems.

Graduate Teaching Assistant, UNH, Durham, NH 2015 – 2017
Led four semesters of independent instruction for introductory biology focusing on evolution, biodiversity, and ecology through inquiry learning.

Manuscripts and Publications

Stewart, Anthony J., et al. “Patterns and drivers of soil carbon after integrating newly revealed wetlands and wet soil carbon stocks”. *In Review* at JGR Biogeosciences. 2026.

Spinola, Diogo et al., (Co-Author **Anthony J. Stewart**). “A Continental-Scale Study of Spodosols across North America and Implications for Soil Organic Carbon Dynamics.” *CATENA* 263 (February 2026): 109743. https://doi.org/10.1016/j.catena.2025.109743.

Stewart, Anthony J., et al. “Revealing the Hidden Carbon in Forested Wetland Soils.” *Nature Communications*. 2024. https://doi.org/10.1038/s41467-024-44888-x.

Halabisky, Meghan, et al., (Co-Author **Anthony J. Stewart**). “The Wetland Intrinsic Potential Tool: Mapping Wetland Intrinsic Potential through Machine Learning of Multi-Scale Remote Sensing Proxies of Wetland Indicators.” *Hydrology and Earth System Sciences* 2023: 3687–99. https://doi.org/10.5194/hess-27-3687-2023.

Campbell, A.D., et al., (Co-Author **Anthony J. Stewart**). “A review of carbon monitoring in wet carbon systems using remote sensing”. Environ. Res. Lett. 2022. https://doi.org/10.1088/1748-9326/ac4d4d

Stewart, Anthony J, et al. “Forest Conversion to Silvopasture and Open Pasture: Effects on Soil Hydraulic Properties.” *Agroforestry Systems* https://doi.org/10.1007/s10457-019-00454-9.

Coble, A.P., et al. (Co-Author **Anthony J. Stewart**). Influence of forest-to-silvopasture conversion and drought on components of evapotranspiration. *Agriculture, Ecosystems & Environment*. 2020 https://doi.org/10.1016/j.agee.2020.106916.

Research Grants And Fellowships

Atkinson Postdoctoral Fellowship 2025
Cornell Atkinson Center for Sustainability Postdoctoral Researcher

Graduate Student Completion Fellowship 2025
Funding support for final spring quarter of PhD

Science Graduate Student Research (SCGSR) Program Fellowship, Dept. of Energy 2024
Funding support for dissertation research at Lawrence Livermore National Lab

Assessing the climate change vulnerability of wetland habitats, WA DNR 2023
Funded project to examine climate related impacts to hydrogeomorphic wetlands in WA

Student Research Support Fund, UW SEFS 2022
Financial support for field work in the Hoh Rainforest, WA

Instrumentation Discovery Travel Grant, CUAHSI 2021
Support for travel to Juneau, AK for datalogger, piezometers, and well implementation

Recent Presentations

The Nature Conservancy Peatland Mapping Workshop 2025
Peatland and Forested wetland mapping across Washington

American Geophysical Union Annual Meeting 2024
1. *Mineral-associated drivers of soil carbon stabilization across wetland and upland soils: evaluating redox patterns and radiocarbon age at a watershed scale*
2. *Teal Carbon Phase 2: Scaling Wetland Carbon Accounting and Uncertainty Estimation to Meet Stakeholder Policy and Decision-Making Need*

American Geophysical Union Annual Meeting 2023
1. *Exploring controls on soil carbon stocks across uplands and wetland over a regional climate gradient in the Pacific Northwest using geospatial data*
2. *Cryptic carbon of the Pacific Northwest: improving wetland soil carbon estimation and monitoring with inclusion of hidden forested wetlands*

Society of Wetland Scientists Annual Meeting 2023
Cryptic Carbon of the Pacific Northwest: Improving Wetland Soil Carbon Estimation and Monitoring with Inclusion of Hidden Forested Wetlands

National Association of Wetland Managers - Wetland Mapping Consortium Webinar 2023
Mapping Wetland Probabilities: Tools, Models, and Applications

American Geophysical Union Annual Meeting 2022
Cryptic carbon: wetland identification under perennial forest cover enhances spatially explicit modeling of soil carbon stock

Northwest Indian Fisheries Commission Tribal Habitat Conference 2022
Climate Change Solutions: Carbon Sequestration in Coastal and Forested Wetlands – Improving estimates of wetland soil carbon beneath the forest canopy through a spatially explicit remote sensing approach

Joint Aquatic Sciences Meeting 2022
Improving estimates of wetland soil carbon beneath the forest canopy through a spatially explicit remote sensing approach

American Geophysical Union Fall Meeting 2021
Improving estimates of wetland carbon beneath a forest canopy through a spatially explicit remote sensing approach

North American Carbon Program Annual Meeting 2021

Wetland carbon stocks under forest canopy: a remote sensing approach

Service and Professional Societies

Professional Societies:

American Geophysical Union, Society of Wetland Scientists, National Association of Wetland Managers

Journals:

JGR Biogeosciences, JGR Geophysical Research Letters, Frontiers in Ecology and Evolution: Conservation and Restoration Ecology, Scientific Reports, Agroforestry Systems

Awards

Outstanding Student Presentation, America Geophysical Union 2023

Outstanding Student Presentation, 7th North American Carbon Program Award 2021

Outreach and News

Washington State Has Been Sitting on a Secret Weapon Against Climate Change

Article appears in The Atlantic

The West's wetlands are struggling. Some have been overlooked altogether 2024

Article appears in High Country News

Learning Forest Guest Article 2024

Using Advanced Mapping to Uncover Hidden Carbon Stocks in Forested Wetlands

US Department of Education Upward Bound Program, Juneau, AK 2022

Developed a first-time science experience in the field for high school students, introducing them to soils and wetlands.

Central High School, Independence, OR, Virtual 2021

Presented on non-traditional routes to college and graduate education as a guest speaker.

Alaska Needs Clarity, Accuracy From Its Intellectual Leaders 2020

Opinion Article appears in The Juneau Empire